

Sijie Lin

Address:

Rotman School of Management
University of Toronto
105 St. George St.
Toronto, Ontario
M5S 3E6, Canada

Phone: +1-647-570-3578**Email:** sijie.lin@mail.utoronto.ca**Website:** www.sijie-lin.com

Research Interests: Industrial Organization, Artificial Intelligence, Applied Microeconomics

EDUCATION

Ph.D. in Economics, University of Toronto 2026 (Expected)
Committee: Matthew Mitchell (co-supervisor), Heski Bar-Isaac (co-supervisor)
Avi Goldfarb, Ambarish Chandra

M.A in Economics, University of Toronto 2020

B.A in Economics, Fudan University 2019

WORKING PAPERS

[Learning to Prompt: Human Adaptation in Production with Generative AI](#) (JMP)

[Hiding From Generative AI](#), R&R at **Quantitative Economics**

WORK IN PROGRESS

Knowledge Spillovers in the Diffusion of Generative AI

Do AI Upgrades Increase the Value of Human Adaptation?

Are Human Users Correcting AI-Created Stereotypes?

with Ruiqi Sun (University of Hong Kong) and Siyuan Liu (University of Toronto)

Platforms in Platform

AWARDS AND GRANTS

CREATE for Accelerated Discovery (AccelD) Stipend Award, University of Toronto	2025
TD MDAL Grant, Rotman School of Management, University of Toronto	2025
TD MDAL Grant, Rotman School of Management, University of Toronto	2024
University of Toronto Doctoral Fellowship	2020 - 2025

CONFERENCE PRESENTATIONS

TSE Digital Economics Conference (Toulouse)	2025
North American Summer Meeting of Econometric Society (Nashville)	2024
ESIF Economics and AI+ML Meeting of Econometric Society (Ithaca)	2024
European Economic Association Annual Meeting (Rotterdam)	2024
American Law and Economics Annual Meeting (Ann Arbor)	2024
Canadian Economics Association Annual Meeting (Toronto)	2024

TEACHING EXPERIENCE

Course Instructor: University of Toronto	2023
ECO 316: Game Theory	

Teaching Assistant: University of Toronto	2019 - 2025
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Undergraduate: Industrial Organization; Quantitative Methods in Economics; Topics in Price Theory; Principles of Macroeconomics; Behavioral and Experimental Economics; Labor Economics

MBA: Decision Making with Models & Data; Microeconomics; Python for Business

COMPUTER SKILLS

Python, STATA, MATLAB, LaTeX, Cloud Computing

REFERENCES

Matthew Mitchell
Rotman School of Management
University of Toronto
105 St. George St.
Toronto, Ontario
M5S 3E6, Canada
matthew.mitchell@rotman.utoronto.ca

Heski Bar-Isaac
Rotman School of Management
University of Toronto
105 St. George St.
Toronto, Ontario
M5S 3E6, Canada
heski.bar.isaac@rotman.utoronto.ca

Avi Goldfarb
Rotman School of Management
University of Toronto
105 St. George St.
Toronto, Ontario
M5S 3E6, Canada
avi.goldfarb@rotman.utoronto.ca

Ambarish Chandra
Rotman School of Management
University of Toronto
105 St. George St.
Toronto, Ontario
M5S 3E6, Canada
ambarish.chandra@rotman.utoronto.ca

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Abstracts

Learning to Prompt: Human Adaptation in Production with Generative AI (Job Market Paper)

What is the role of human input in AI-assisted production? Humans interact with language-based AI through prompts, combinations of words. A key feature of *human* input is *adaptation*: users dynamically modify their prompts based on their understanding of AI. In this paper, I empirically investigate two types of adaptation: (1) adaptation to new AI versions, referring to how people change their prompts in response to AI upgrades; (2) adaptation to outputs from previous prompts, referring to how people adjust their prompts iteratively to converge on desired outcomes. I study this adaptation using prompt-level data from Midjourney, a leading AI image generator. First, I show that users adapt to AI upgrades by writing different words in their prompts. By submitting prompts written for the old version to the new and vice versa, I decompose the output shifts as arising from prompt changes (73%), AI changes (20%), and an AI-human cross effect (7%), implying strong complementarity between AI and human inputs. Second, I study the role of adaptation to previous output by analyzing how prompts evolve within the creative process of an artwork. I estimate a structural model of the creative process using the sequential search framework. Users judge the image quality of the preceding prompt to inform the next prompt. In counterfactual analysis, I show that when users do not adjust prompts in accordance with such judgment, they need three times as many prompts in each session to reach the same outcome as observed in data. Both results highlight the importance of human judgment and adaptation in the creative process.

Hiding From Generative AI

How does generative Artificial Intelligence (AI) change the behaviors of content creators? I investigate the effect of an AI image generator on artists' incentives to publish artworks using data from an online art platform, DeviantArt. On November 11 2022, DeviantArt introduced a generative AI image generator into the platform and artworks on this platform entered training data by default. Using a difference-in-differences estimation with artists who do not use AI, I show that digital artists publish 21% fewer artworks following AI's introduction on this platform, in contrast to artisan crafts artists. This reduction could potentially hinder knowledge spillovers to other artists and AI training data availability. By matching the artworks of artists who publish both on DeviantArt and Instagram, I find that despite artists publishing fewer artworks on DeviantArt, the quality of published artworks for a given artist remains the same after the introduction of AI.

Platforms in Platform

Platforms could be part of a larger platform like an app store. Even though there exists a positive cross-group network externality between buyers and sellers, there is still an incentive for buyers and sellers to segment into different platforms. Buyers may choose to use different platforms to access more targeted products, while sellers may use different platforms to avoid competition for buyers' attention. This paper argues that segmenting the market using smaller platforms can be profitable for the large platform, as it can charge different platform fees to sellers of different quality and save buyers' attention to better sellers. This paper also shows that when market agglomeration is more profitable for the large platform, it may choose to ban smaller platforms, which could be socially inefficient as it may deter mediocre sellers from entering the market.